SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct

I Nevitha Magendran/192521206 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Nevitha

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

   Answer:

   Auto-scaling: Auto-scaling automatically adds or removes virtual machine / container instances according to rules such as CPU utilisation above 70%, request count or scheduled events. Before a seasonal sale starts, a scheduled policy can pre-warm extra

capacity; during an unexpected spike, a dynamic (target-tracking) policy launches more instances within minutes; after the rush, surplus instances are terminated. The company therefore pays only for what it uses and never has to buy hardware for peak load.

Load balancing: A load balancer (e.g., AWS Elastic Load Balancer, Azure Load Balancer) is the single entry point for users. It distributes incoming requests across all healthy instances using algorithms such as round robin or least connections, performs health checks and stops sending traffic to failed servers. As auto-scaling adds new instances, they are registered with the load balancer automatically, so the extra capacity is used immediately.

Reducing downtime and improving user experience: No single server becomes a bottleneck or a single point of failure. Instances are spread across multiple Availability Zones, so even if one zone fails, traffic is rerouted to the others. Users get fast page loads, working carts and successful payments even at peak time, which lowers cart abandonment and protects revenue and brand reputation.

Role of monitoring tools: Tools such as Amazon CloudWatch, Azure Monitor or Prometheus/Grafana collect metrics (CPU, memory, latency, error rate, request count) and logs. These metrics trigger the scaling policies, raise alarms to the operations team, show dashboards in real time and help in capacity planning and cost analysis after the sale.

Real-world scenario: During an online festival sale (such as Big Billion Days or Black Friday), an e-commerce site's traffic may increase 10 times within minutes. With auto-scaling groups behind an Application Load Balancer and CloudWatch alarms, the web tier grows from 10 to 100 instances automatically, the site stays responsive, and after the sale it shrinks back to normal, saving cost.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data?

Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Answer:

Durability and data redundancy: Cloud object storage (e.g., Amazon S3, Azure Blob, Google Cloud Storage) automatically stores multiple copies of each object across different devices and Availability Zones, and can replicate data to other regions (cross-region replication). Providers advertise very high durability (for example, S3 offers 99.999999999% – "eleven nines"). Checksums detect corrupted data and repair it from the other copies, and versioning protects against accidental deletion or overwriting.

Encryption and access control: Data is encrypted in transit (TLS/HTTPS) and at rest (AES-256) using provider-managed or customer-managed keys (KMS / HSM). Identity and Access Management (IAM) policies, role-based access control, multi-factor authentication, bucket policies and private network endpoints ensure that only authorised users and applications can read or modify the backups. Object lock / immutability protects the backups from ransomware, and access logs provide an audit trail.

Cost advantages: There is no upfront capital expense for disks, tape libraries, data-centre space, cooling or power, and no staff needed to maintain them. The pay-as-you-go model charges only for the storage used, capacity is virtually unlimited, and lifecycle policies move older backups to cheaper tiers (e.g., S3 Glacier / Azure Archive), which cost a fraction of the price of traditional storage. Replication and hardware refresh are handled by the provider.

Disaster recovery example: A hospital or bank backs up its databases every night to cloud storage replicated in a second region. If a fire, flood or ransomware attack destroys the primary data centre, the organization launches its servers in the secondary region, restores the latest backup and resumes operations within minutes or hours (low RTO and RPO) instead of days.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Answer:

How CDNs improve performance and reduce latency: A Content Delivery Network is a globally distributed network of servers that caches copies of content (images, videos, scripts, API responses) close to users. Instead of every request travelling to the single origin server, it is served from the nearest cache. This reduces the physical distance and number of network hops, so latency, load time and bandwidth cost fall while the load on the origin server also drops.

Edge servers and user experience: Edge servers (points of presence) located in many cities and countries deliver content locally. A user in India, Europe or the USA receives data from a nearby node, giving faster app start-up, smoother video playback, fewer buffering events and better reliability. Edge nodes also absorb traffic spikes and DDoS attacks, can terminate TLS, compress content and even run lightweight code (edge computing).

Global availability by cloud providers: Cloud providers operate multiple regions and Availability Zones worldwide. The company can deploy the application in several regions, use global load balancing / DNS routing (e.g., Route 53, Traffic Manager) to send users to the nearest healthy region, replicate databases across regions and fail over automatically if a region goes down. Providers also supply CDNs such as CloudFront, Azure CDN or Cloud CDN.

Streaming / gaming example: A video-streaming service such as Netflix or YouTube stores popular videos on edge servers (CDN caches) near viewers, so a user gets HD video with little buffering. In an online multiplayer game, game patches, maps and assets are distributed through the CDN, and players are connected to the nearest regional game server, which keeps ping low and gameplay smooth for players worldwide.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

   Answer:

   Regulatory compliance and auditing: Major cloud providers hold independent certifications and attestations such as ISO 27001, SOC 1/2/3, PCI-DSS, GDPR and HIPAA, and publish compliance reports (e.g., AWS Artifact, Azure Compliance Manager). Under the shared responsibility model the provider secures the infrastructure while the bank secures its own data, configuration and access. Tools such as AWS Config, Azure Policy and CloudTrail continuously check configurations and record every action, making audits easier.

   Role of logging, monitoring and encryption: Logging (CloudTrail, Azure Activity Log) records who did what and when, which supports audit trails and forensic investigation. Monitoring and SIEM tools (CloudWatch, Azure Sentinel, GuardDuty) detect unusual activity such as failed log-ins or unexpected data transfers and raise real-time alerts. Encryption protects data at rest (AES-256) and in transit (TLS), with keys managed in KMS/HSM so that stolen data is unreadable.

   Ensuring data sovereignty: The company should choose cloud regions located inside the country or legal jurisdiction required (for example, Indian regions for data that must stay in India), use policies that deny creation of resources in other regions, disable cross-region replication, keep encryption keys under its own control (customer-managed or hold-your-own-key), and sign contracts / data-residency commitments with the provider. Private connectivity and regular audits confirm that the data never leaves the permitted location.

   Example: A bank runs its online-payment and UPI transaction system on the cloud. Card numbers and account details are encrypted, every transaction is logged, an

anomaly-detection tool flags suspicious transfers (possible fraud) instantly, access is limited to authorised staff using MFA and least-privilege roles, and all data is stored only in the in-country region to satisfy regulator rules such as RBI data-localisation requirements.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

   Answer:

   Containers vs. virtual machines: A virtual machine includes a full guest operating system on top of a hypervisor, so each VM needs gigabytes of storage and memory and takes minutes to boot. A container packages only the application and its libraries and shares the host operating-system kernel, so it is lightweight (megabytes), starts in seconds, has near-native performance and allows many more containers than VMs to run on the same hardware. VMs give stronger isolation; containers give higher density and speed.

   Orchestration with Kubernetes: Kubernetes automates deployment, scaling and management of containers. It schedules containers on suitable nodes, restarts failed ones (self-healing), scales pods up or down automatically (Horizontal Pod Autoscaler), balances load through services, performs rolling updates and rollbacks without downtime, and manages configuration and secrets. The team describes the desired state in YAML files and Kubernetes maintains it.

   Portability: A container image runs the same way on a developer laptop, an on-premises server or any cloud (AWS EKS, Azure AKS, Google GKE), because it carries all its dependencies (“build once, run anywhere”). This avoids “works on my machine” problems, simplifies CI/CD, enables hybrid and multi-cloud deployment and reduces vendor lock-in.

Microservices example: An online food-delivery or shopping application is split into independent microservices — User/Login, Product Catalogue, Cart, Order, Payment and Notification — each packaged in its own container, with its own database and communicating through REST APIs. Kubernetes can scale the Order and Payment services during peak hours without scaling the others, and one service can be updated or fail without bringing down the entire application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity